

LUKE BANICEVIC

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Experience

ResMed

Dec. 2021 – Present

Software Engineer – Product Development

Bella Vista, Sydney

Skills: *C++, Python, Java, BLE/Bluetooth, Cellular (LTE/2G/3G), TLS/PKI, REST APIs, AWS EKS/Kubernetes, Datadog, Java Profiling (JFR/JVM), SQL, Snowflake, Docker, CI/CD*

- Implemented cellular connectivity in **C++** including AT command handling, IPv4/v6 network stack integration, and connectivity state machines; led IPv4/v6 certification testing on a Rohde & Schwarz LTE callbox and delivered over-the-air firmware upgrade support.
- Engineered a **BLE** RPC data pathway in **C++**, enabling the patient smartphone app to act as an HTTPS proxy for cloud communication from the device, and implemented firmware-side over-the-air upgrade delivery over this channel.
- Designed end-to-end BLE-to-cloud device registration testing across embedded firmware, a patient app emulator, and physical iOS devices, running UAT against a **microservice deployed on AWS EKS/Kubernetes** and using **Datadog** to trace requests and surface **TLS/PKI** certificate and authentication blockers before production rollout.
- Built device provisioning tooling in **Python** integrating the Vodafone **REST API** for IMSI (SIM identity) management and automated SIM profile switching, eliminating 5 manual setup steps and enabling repeatable environment switching across development teams.
- Diagnosed EU SMS delivery failures in device configuration, tracing the fault with Vodafone and Telit across modem firmware, SMSC (carrier message routing), and backend APIs.
- Contributed to backend **Java** microservices on the cloud platform team, shipping new deployments to **AWS EKS/Kubernetes** and diagnosing a production memory leak through **Java profiling (JFR/JVM)** and **Datadog** pod observability across regional deployments.
- Traced fleet-wide device issues by assembling forensic traces of individual devices and regional device groups through **SQL** queries against **Snowflake**, isolating root causes of field problems across a fleet of 5M+ devices spanning the APAC, AMR, EU, and CN regions.

UNSW Solar Racing Team – Sunswift Racing

Feb. 2020 – Dec. 2021

Embedded Software Engineer – Embedded Systems Team

Kensington, Sydney

Skills: *C, CAN Bus, Embedded Systems*

- Designed and built **embedded systems** for UNSW's Sunswift solar car, competing in the 2021 Bridgestone World Solar Challenge.
- Programmed **CAN bus** nodes for integration with vehicle drive-by-wire systems, delivering reliable communication protocols in **C**.

Projects

AI-Assisted CVE Applicability Pipeline

Skills: *Python, LLMs, Grype, Embedded Linux*

- Designed a 3-tier matching engine in **Python** for an **embedded Linux** medical device to automate Linux kernel CVE triage, combining deterministic rule filtering with **LLM** analysis and **Grype** scanning for ambiguous cases, triaging 600+ kernel CVEs and saving ~50 hours of previously manual security review.

URLabs – Dynamic QR Code Platform

Skills: *TypeScript, Next.js, React, Supabase (Postgres, Auth, RLS), Tailwind CSS, Vercel*

- Built and shipped a full-stack **Next.js/React** SaaS for dynamic QR codes that repoint to any destination without reprinting, on a **Supabase Postgres** backend secured by row-level security and server-enforced per-plan limits.
- Designed the scan-capture redirect pathway and per-code analytics dashboard, backed by a nightly **Vercel** cron job that rolls up scan statistics, and shipped behind CI gates running Jest, Playwright, and type checks.

Education

University of New South Wales (UNSW)

Feb. 2018 – Jan. 2023

B.E Computer Science, B.E (Hons) Mechatronic Engineering

Kensington, Sydney